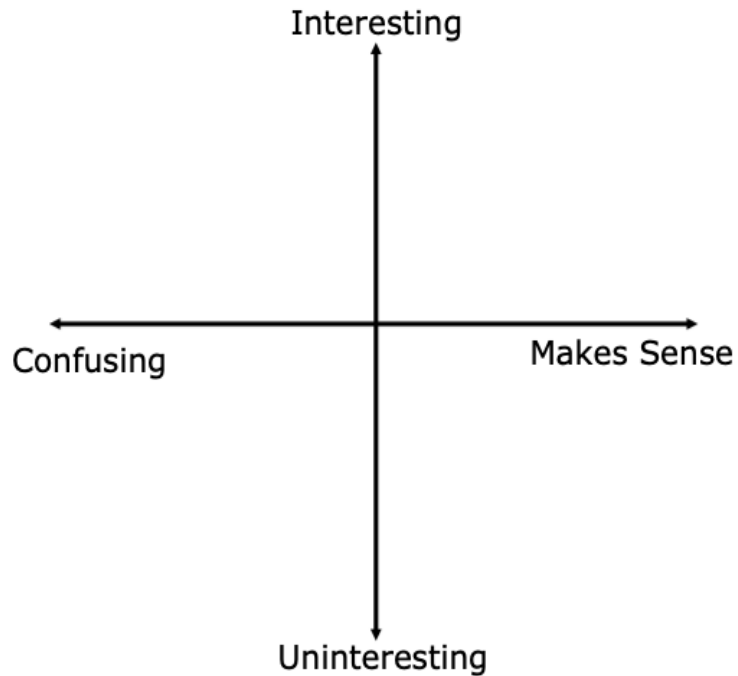




6.H1.4 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest and understanding for today's learning target.



Explain the location of your point.

Unit 6, Lesson H2: Proofs by Contradiction



Benchmark

MA.912.LT.4.8 Construct proofs, including proofs by contradiction.



Learning Target

- ☐ I can use proof by contradiction to justify that all circles are similar.



6.H2.1 Warm-Up: Critical Thinking

1. DirecTV commercials are entertaining because the conditional statements are outrageous. Write five conditional statements with an outrageous conclusion that could be used for a DirecTV commercial.



6.H2.2 Exploration: Contradictions

In the DirecTV commercial, there were seven conditional statements before the final conclusion.

Order	Statement
1	If your cable company puts you on hold, then you get angry.
2	If you get angry, then you go blow off steam.
3	If you go blow off steam, then an accident happens.
4	If an accident happens, then you get an eye patch from the doctor.
5	If you get an eye patch, then people think you are tough.
6	If people think you are tough, then they want to fight you.
7	If people want to fight you, then you get hurt.
8	You will end up in a roadside ditch.

1. Assume that the conclusion is not true. Write a contradiction to the conclusion.
2. Share your contradiction with your partner.

3. Use your contradiction to write a “rewind” for the commercial with statements that would lead to the contradiction.

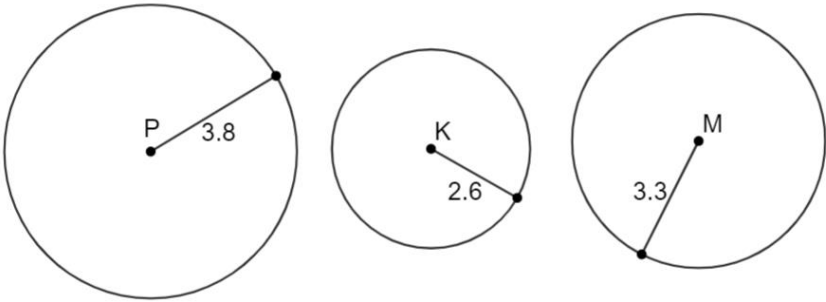
Order	Statement
8	
7	
6	
5	
4	
3	
2	
1	

4. Explain how writing the contradiction would change the commercial.



6.H2.3 Exploration: Circles Are Similar

1. Circles P , K , and M are shown. The radius of Circle P is 3.8 units long. The radius of Circle K is 2.6 units long. The radius of Circle M is 3.3 units long.



a. With your partner, fill in the table. Round to nearest thousandth. Use the pi button on your calculator.

Circle	Radius	Circumference

b. Complete the table.

Circle P and K	
Ratio of Radius of Circle P and Radius of Circle K	
Ratio of Circumference of Circle P and Circumference of Circle K	

c. What do you notice about the ratios?

d. Complete the table.

Circle K and M	
Ratio of Radius of Circle K and Radius of Circle M	
Ratio of Circumference of Circle K and Circumference of Circle M	

e. What do you notice about the ratios?

2. Make a conjecture about the ratios of radii and circumferences of circles.



6.H2.4 Guided Instruction: Proof by Contradiction

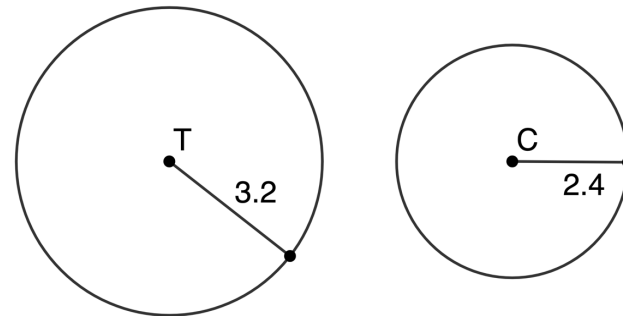
A way to write a proof is by using a contradiction.



Proof by Contradiction

Assume the statement to be proven is false, and work to show its falsity until the result of that assumption is a contradiction.

1. Circles T and C are shown. The radius of Circle T is 3.2 units long. The radius of Circle C is 2.4 units long.



Prove that Circle T is similar to Circle C .

a. Write an assumption that the statement is false.

b. Find the ratio of the radius of Circle T to the radius of Circle C .

- c. Find the ratio of the circumference of Circle T to the circumference of Circle C .

- d. Explain how the ratios show the assumption is not true.

- e. Complete the proof.

Assume that Circle T is _____ to Circle C . The ratio of the radius of Circle T to the radius of Circle C is _____. The ratio of the circumference of Circle T to the circumference of Circle C is _____. The ratio of the radii and the ratio of the circumferences are _____, so corresponding parts of Circle T and Circle C are proportional. Since the corresponding parts of Circle T and Circle C are proportional, the circles are _____. This is a contradiction, so the assumption that Circle T is not similar to Circle C is not true.



6.H2.5 Wrap-Up: Reflection

1. Check below all the statements that apply to your understanding.
 - ☐ I understand how to write a contradiction.
 - ☐ I need support on how to write a contradiction.
 - ☐ I understand that all circles are similar.
 - ☐ I need support on understanding why all circles are similar.
 - ☐ I understand how to prove that all circles are similar by contradiction.
 - ☐ I need support on how to prove that all circles are similar by contradiction.